

# Introduction to Earth System

## Extreme events

Hydrometeorological hazards, climate-extreme indices, and extreme-value statistics

# 1

## Hydrometeorological hazards & Extremes

Atmospheric processes that generate damaging water, wind and temperature extremes

# Hydrometeorological hazards

## Convective storms

thunderstorm, hail, lightning,  
tornado, derecho

## Tropical cyclones

warm-core vortex, heavy rain,  
wind, storm surge

## Monsoon & frontal rain

persistent ascent, moisture  
convergence, floods

## Temperature extremes

heat wave, cold wave, frost,  
wildfire weather

## Drought

precipitation deficit + high  
evaporative demand

## Compound events

co-occurring or sequential drivers  
and impacts

A hazard becomes **a disaster** only through exposure and vulnerability; the physics determines the forcing, not the social loss by itself.

### Example:

- A. 160 mm of rain in 3 hours over an urban catchment
- B. Seven consecutive days with daily Tmax above the local 90th percentile
- C. Landfalling typhoon with wind, surge, and inland rainfall
- D. Thirty days with RR < 1 mm during the rainy season

# Compound extremes are multivariate tail events

$$P(X > x, Y > y) = P(X > x) \cdot P(Y > y \mid X > x)$$

## Hot + dry

heat stress, wildfire weather, crop losses

## Wind + surge + rain

cyclone coastal compound flooding

## Sequential storms

catchment memory increases flood risk

## Hot nights + humidity

human thermoregulation stress

# Observed losses

1970-2021 reported weather, climate and water-related disasters

**11,778**

reported disasters

**>2 million**

reported deaths

**US\$4.3T**

economic losses

- Risk analysis must separate hazard probability from exposure, vulnerability, and reporting changes.
- Early warning can reduce mortality even while economic exposure grows.

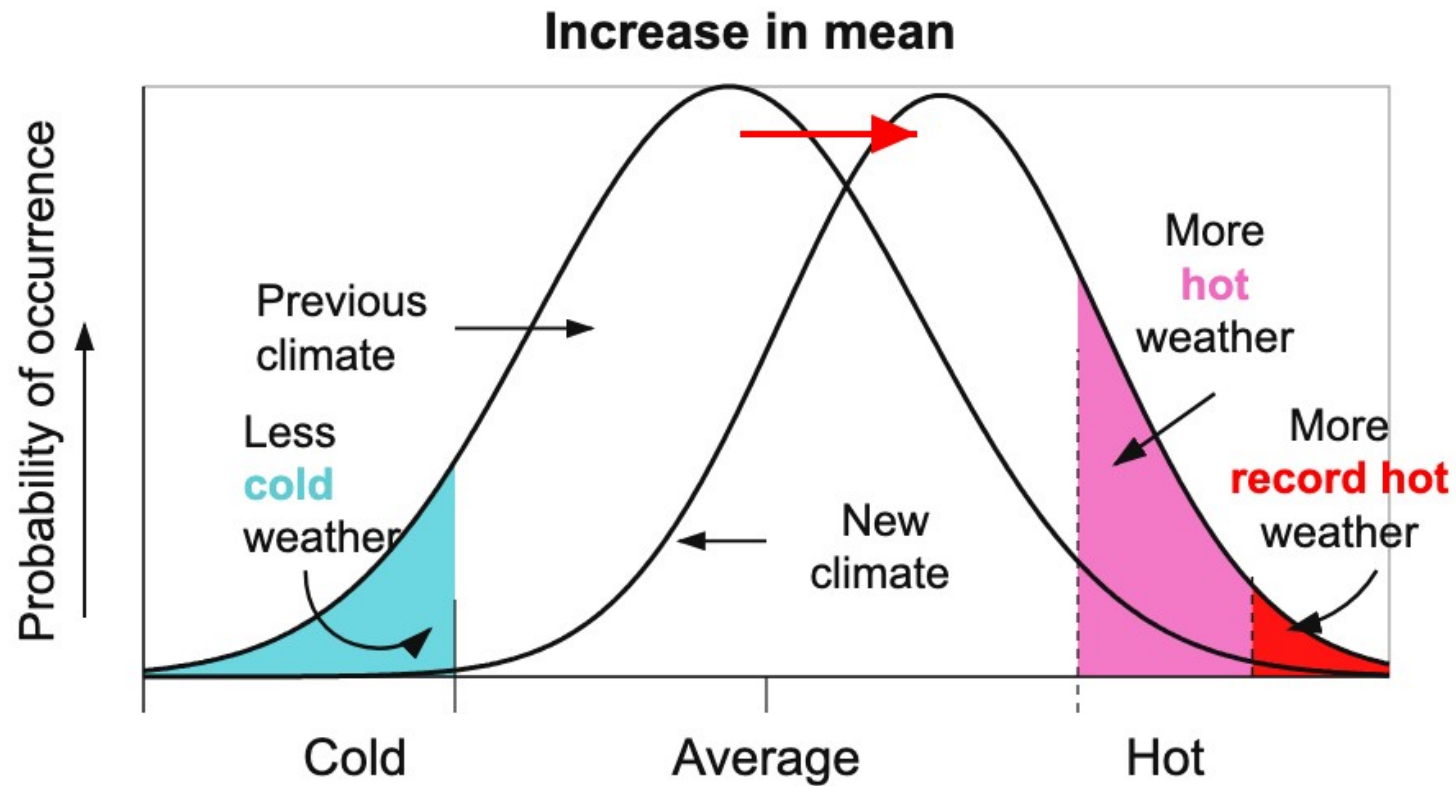
# 2

## Definitions and climate-extreme indices

# IPCC definition: extreme is relative to place, season and distribution

Extreme weather event: rare at a particular place and time of year; often  $\leq P10$  or  $\geq P90$  of an observed probability distribution.

- The same absolute temperature can be ordinary in one climate and extreme in another.
- The reference distribution must specify variable, location, season, time scale, and baseline period.
- Climate extreme can refer to weather events or longer climate variables such as seasonal drought.

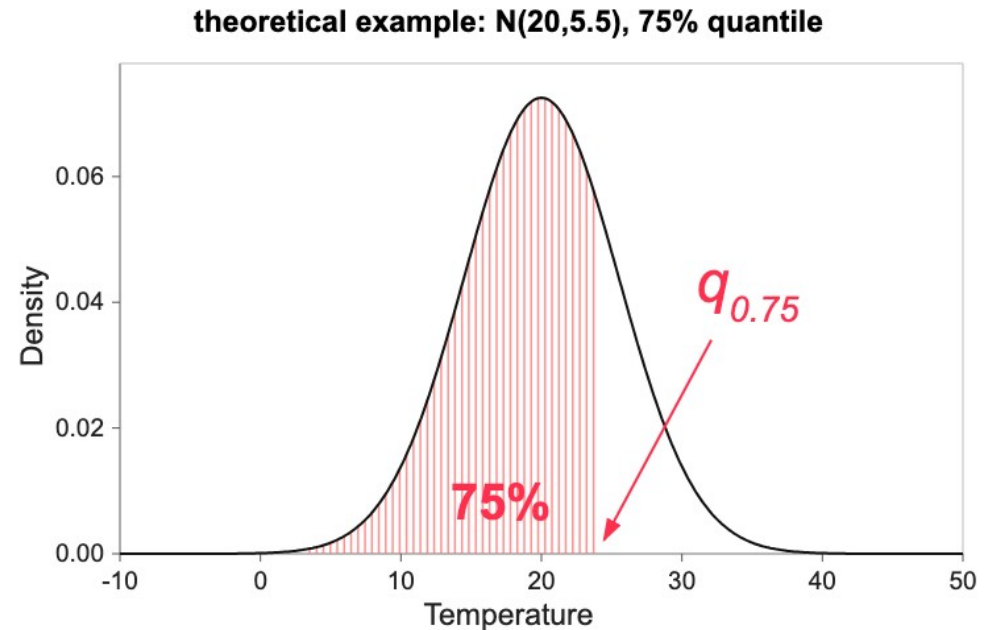


IPCC – Third Assessment Report (TAR)

# Percentile

## Definition:

The  $P^{\text{th}}$  **percentile** of a list of  $N$  ordered values (sorted from least to greatest) is the smallest value in the list such that  $P$  percent of the data is less than or equal to that value.



**Exercise #1:** Given the daily rainfall values of 5 consecutive days {10,5, 25,40,30}.

What are the 30<sup>th</sup>, 40<sup>th</sup>, 50<sup>th</sup>, and 100<sup>th</sup> of the list?

*Hint: The Nearest Rank Method:*  $n = \lceil P \times N / 100 + 0.5 \rceil$

**Python:** `np.percentile(a, p)`  
a: array of values  
p: percentile, eg. 10

**Exercise #1:** Given the daily rainfall values of 5 consecutive days: {10,5, 25,40,30}. What are the 30<sup>th</sup>, 40<sup>th</sup>, 50<sup>th</sup>, and 100<sup>th</sup> of the list?

**Solution:**

- 1) Order the data from least to greatest → {5, 10, 25, 30, 40}
- 2) For each percentile, calculate the ordinal rank n

$$n = [P \times N / 100 + 0.5]$$

- 3) The value at the n<sup>th</sup> position in the sorted list is the P<sup>th</sup> percentile

**30th Percentile (P=30)**

- Calculate rank (n):

$$n = [30 \times 5 / 100 + 0.5] = [2] = 2$$

- The 2<sup>nd</sup> value in the list is 10.

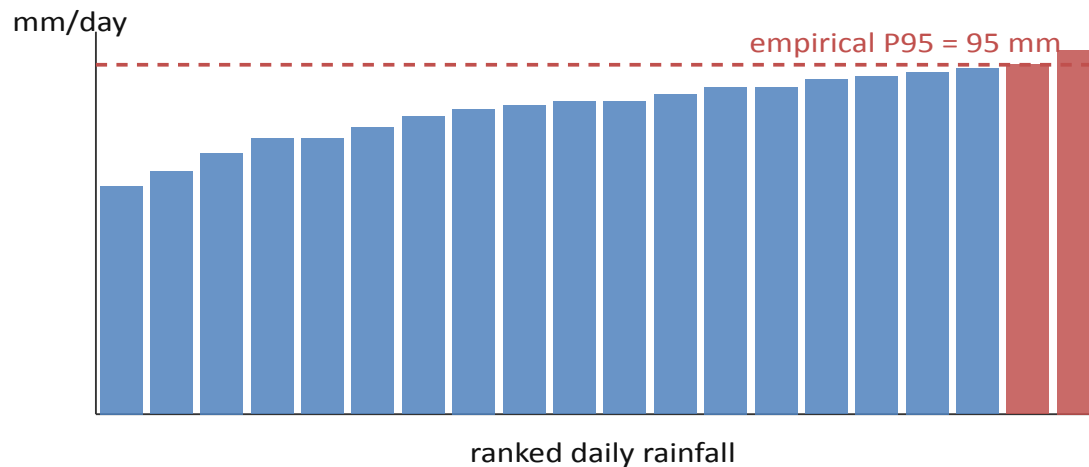
→ 30th percentile = 10

## Exercise 2: percentile threshold for daily rainfall

Given daily rainfall values (mm/day) for 20 days:

66, 62, 78, 75, 99, 71, 81, 83, 84, 89, 85, 87, 85, 89, 91, 92, 93, 94, 96, 75

- 1. Find the 95<sup>th</sup> percentile using the nearest-rank method.
- 2. Find the percentile rank of a rainfall value of 85 mm/day.
- 3. Which days would be counted in R95p (values  $\geq$  the 95<sup>th</sup> percentile)?



## Solution:

### 1) the 95<sup>th</sup> percentile?

- Sort the data
- 95<sup>th</sup> percentile  $\rightarrow$  the ordinal rank  $n = [P \times N / 100 + 0.5] = [95 \times 20 / 100 + 0.5] = 19$
- The 19<sup>th</sup> value in the sorted list is 95  $\rightarrow$  the 95<sup>th</sup> percentile is 96 mm/day

### 2) Find the percentile rank of a rainfall value of 85 mm/day?

- 85 mm/day has the ordinal rank  $n=10$  and 11.
- $n = [P \times N / 100 + 0.5] \rightarrow 10 = [P \times N / 100 + 0.5] \text{ \& } 11 = [P \times N / 100 + 0.5]$
- $\rightarrow P \times N / 100 - 0.5 < 10 \leq P \times N / 100 + 0.5 \rightarrow 47.5 \leq P < 52.5$ 
  - $\& P \times N / 100 - 0.5 < 11 \leq P \times N / 100 + 0.5 \rightarrow 52.5 \leq P < 57.5$

$\rightarrow 47.5 \leq P < 57.5$

### 3) Days to be counted in R95p $\rightarrow$ days with values of 96 mm & 99 mm

## Climate-extreme index (CEI)

- The CEI was first introduced in early 1996 (Karl et al. 1996)
- General lack of observational data on extremes
- Provided a means to communicate with policy makers regarding our understanding of changes in climate
- Originally looked at historical data (1+ years old) on an annual basis

**ETCCDI** (The Expert Team on Climate Change Detection and Indices)

<https://www.wcrp-climate.org/etccdi>

## ETCCDI temperature indices translate tails into counts

**TX90p**

**Warm days**

% of days when TX > calendar-day 90th percentile

**TN10p**

**Cool nights**

% of days when TN < calendar-day 10th percentile

**WSDI**

**Warm spell duration**

Count of days in spells of at least (3) 6 days with TX > P90

**CSDI**

**Cold spell duration**

Count of days in spells of at least (3) 6 days with TN < P10

TX: maximum daily temperature

TN: minimum daily temperature

# ETCCDI precipitation indices separate intensity, frequency and persistence

**RX1day**

Maximum 1-day precipitation in a period

**RX5day**

Maximum consecutive 5-day precipitation total

**R95p/R99p**

Annual total precipitation from very/extremely wet days

**SDII**

Mean precipitation intensity on wet days:  $\text{PRCPTOT} / \text{wet-day count}$

**CDD/CWD**

Maximum number of consecutive dry/wet days

Wet day: days with  $R \geq 1\text{mm}$

PRCPTOT: annual total rainfall

# Heatwave definitions mix percentile thresholds and spell logic

Heatwave day =  $[TX_d > q90\_TX(d)]$  OR  $[TN_d > q90\_TN(d)]$  for at least  $m$  consecutive days

- Absolute threshold: useful for impacts, e.g., days above 35 °C.
- Percentile threshold: useful for comparing different climates.
- Spell threshold: captures persistence, nighttime recovery, and accumulated heat stress.

## Exercise 3: compute TX90p for a short target year

**Baseline calendar-day P90 thresholds for 10 days:**

32, 33, 34, 34, 35, 35, 34, 33, 33, 32 °C

**Observed TX in target year:**

31, 34, 35, 36, 34, 36, 37, 32, 35, 33 °C

- Count days where TX > baseline P90.
- Compute  $\text{TX90p} = 100 * \text{count} / \text{total valid days}$ .
- Identify the longest warm spell if a spell requires at least 3 consecutive exceedance days.

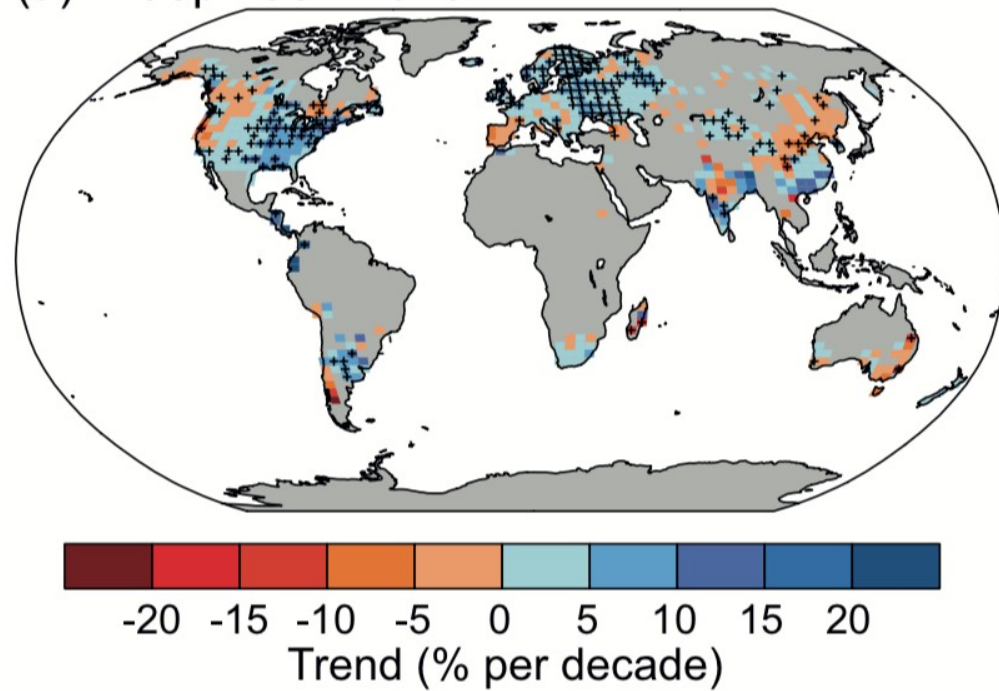
### Solution 3: TX90p and warm spell

$$I_d = 1[TX_d > P90_d] = 0, 1, 1, 1, 0, 1, 1, 0, 1, 1$$

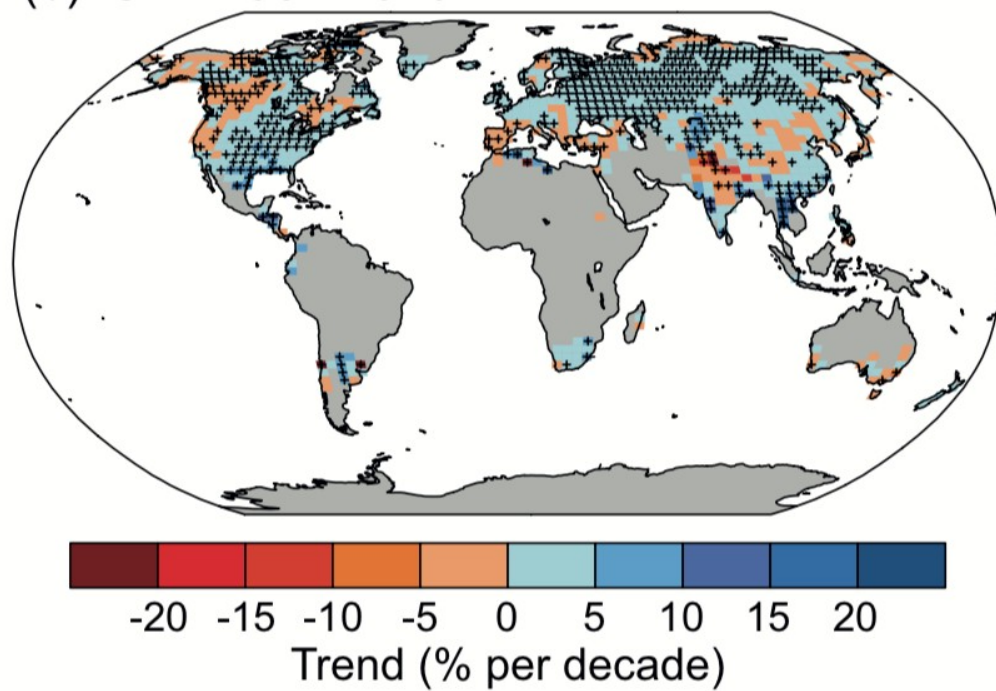
$$TX90p = 100 \times (7/10) = 70\%$$

- Warm spells: days 2–4 length 3; days 6–7 length 2; days 9–10 length 2.
- the longest warm spell → days 2–4

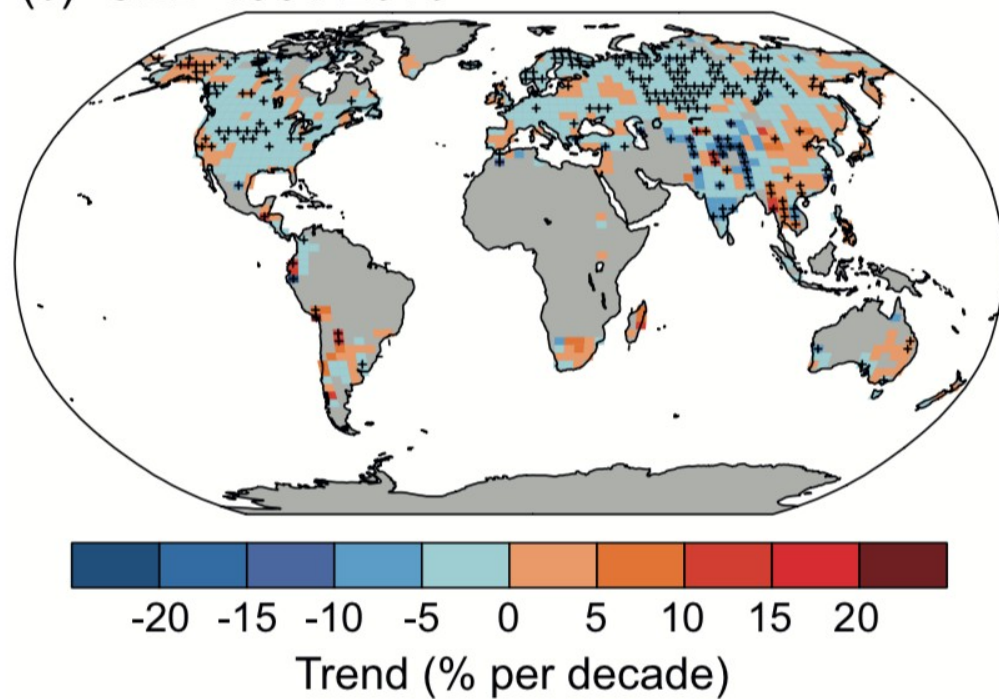
(a) R95p 1951-2010



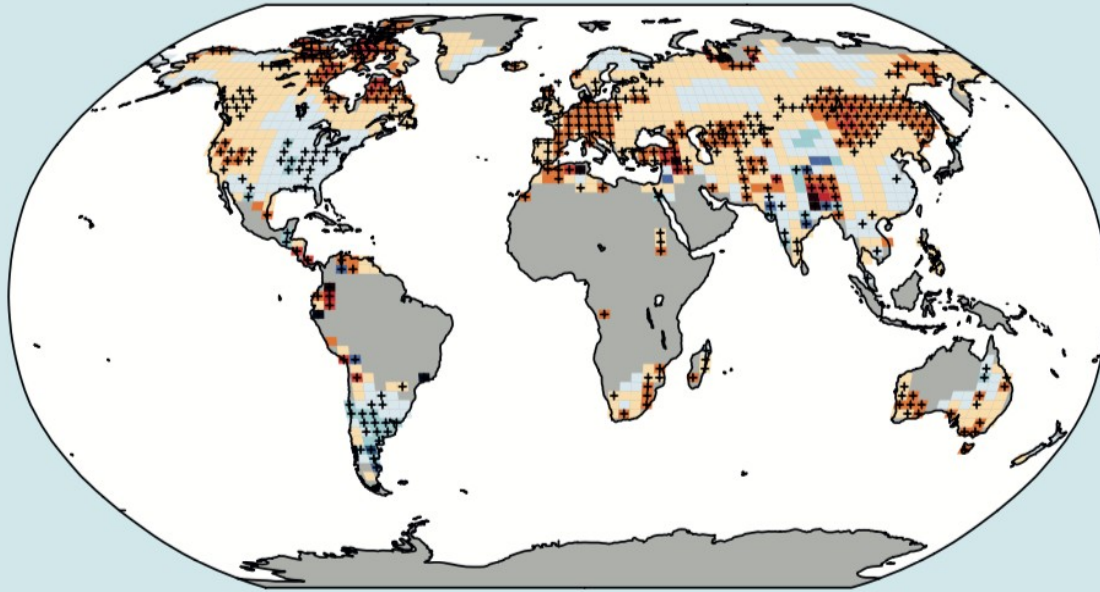
(b) SDII 1951-2010



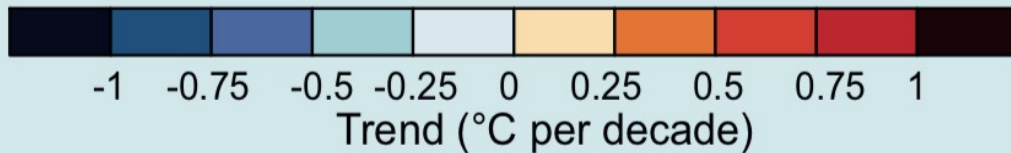
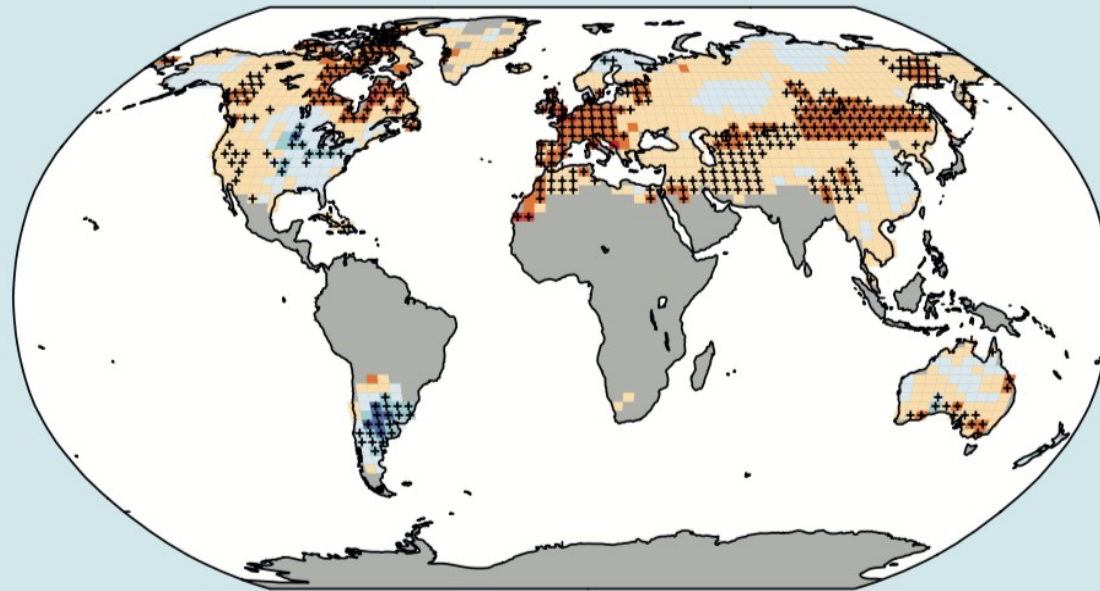
(c) CDD 1951-2010



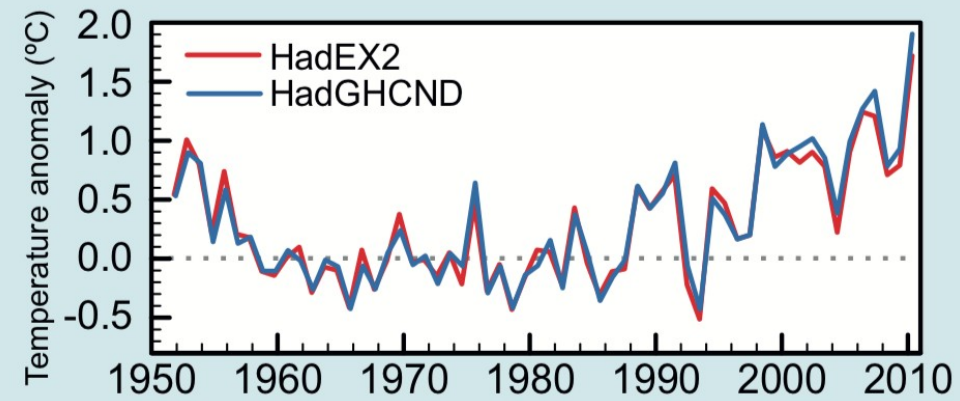
(a) HadEX2 1951-2010



(b) HadGHCND 1951-2010



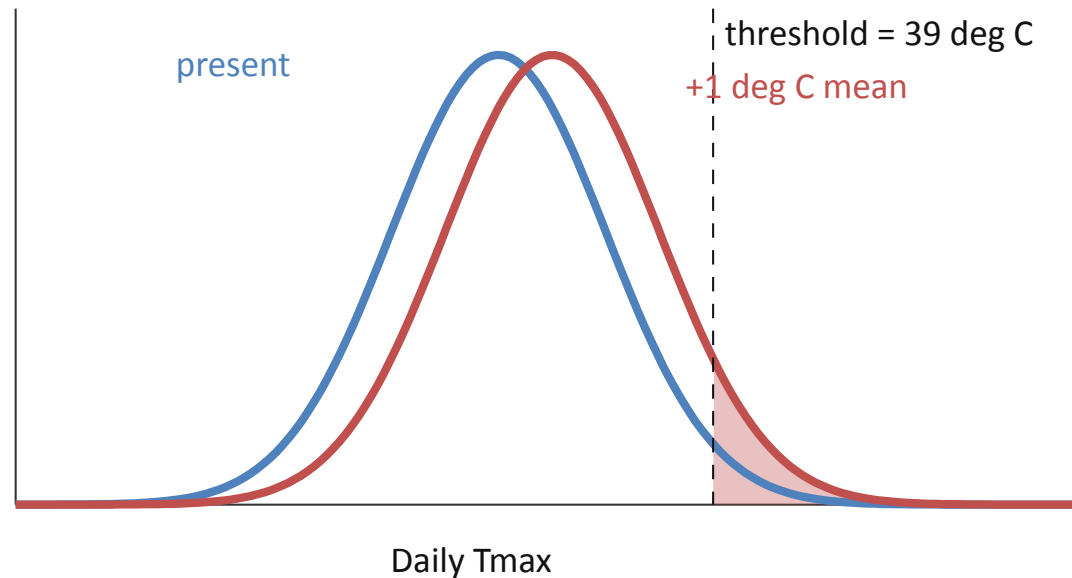
(c) Global land average



IPCC AR5

Trend of TXx during 1951-2010

## A small mean shift can produce a large tail-risk shift



- For Gaussian-like central variability, extremes depend exponentially on the squared standardized threshold.
- A 1 deg C shift is small near the mean but large if the threshold is 2-3 standard deviations away.

## Exercise 4: probability ratio for a shifted temperature distribution

Present:  $X \sim N(35, 2^2)$ . Heat threshold = 39 °C.

Warmed climate:  $X' \sim N(36, 2^2)$ . Same threshold.

- 1. Compute z-scores for both climates.
- 2. Estimate  $P(X > 39)$  and  $P(X' > 39)$ .
- 3. Compute the probability ratio  $PR = P_{\text{warm}} / P_{\text{present}}$ .

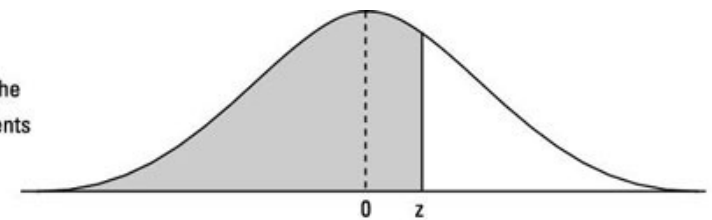
## Solution 4: a 1 deg C shift almost triples exceedance probability

Present  $z = (39 - 35)/2 = 2.0 \Rightarrow P = 0.0228$

Warmed  $z = (39 - 36)/2 = 1.5 \Rightarrow P = 0.0668$

$PR = 0.0668 / 0.0228 = 2.94$

Number in the  
table represents  
 $P(Z \leq z)$



| z   | 0.00  | 0.01  | 0.02  | 0.03  | 0.04  | 0.05  | 0.06  | 0.07  | 0.08  | 0.09  |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 0.0 | .5000 | .5040 | .5080 | .5120 | .5160 | .5199 | .5239 | .5279 | .5319 | .5359 |
| 0.1 | .5398 | .5438 | .5478 | .5517 | .5557 | .5596 | .5636 | .5675 | .5714 | .5753 |
| 0.2 | .5793 | .5832 | .5871 | .5910 | .5948 | .5987 | .6026 | .6064 | .6103 | .6141 |
| 0.3 | .6179 | .6217 | .6255 | .6293 | .6331 | .6368 | .6406 | .6443 | .6480 | .6517 |
| 0.4 | .6554 | .6591 | .6628 | .6664 | .6700 | .6736 | .6772 | .6808 | .6844 | .6879 |
| 0.5 | .6915 | .6950 | .6985 | .7019 | .7054 | .7088 | .7123 | .7157 | .7190 | .7224 |
| 0.6 | .7257 | .7291 | .7324 | .7357 | .7389 | .7422 | .7454 | .7486 | .7517 | .7549 |
| 0.7 | .7580 | .7611 | .7642 | .7673 | .7704 | .7734 | .7764 | .7794 | .7823 | .7852 |
| 0.8 | .7881 | .7910 | .7939 | .7967 | .7995 | .8023 | .8051 | .8078 | .8106 | .8133 |
| 0.9 | .8159 | .8186 | .8212 | .8238 | .8264 | .8289 | .8315 | .8340 | .8365 | .8389 |
| 1.0 | .8413 | .8438 | .8461 | .8485 | .8508 | .8531 | .8554 | .8577 | .8599 | .8621 |
| 1.1 | .8643 | .8665 | .8686 | .8708 | .8729 | .8749 | .8770 | .8790 | .8810 | .8830 |
| 1.2 | .8849 | .8869 | .8888 | .8907 | .8925 | .8944 | .8962 | .8980 | .8997 | .9015 |
| 1.3 | .9032 | .9049 | .9066 | .9082 | .9099 | .9115 | .9131 | .9147 | .9162 | .9177 |
| 1.4 | .9192 | .9207 | .9222 | .9236 | .9251 | .9265 | .9279 | .9292 | .9306 | .9319 |
| 1.5 | .9332 | .9345 | .9357 | .9370 | .9382 | .9394 | .9406 | .9418 | .9429 | .9441 |
| 1.6 | .9452 | .9463 | .9474 | .9484 | .9495 | .9505 | .9515 | .9525 | .9535 | .9545 |
| 1.7 | .9554 | .9564 | .9573 | .9582 | .9591 | .9599 | .9608 | .9616 | .9625 | .9633 |
| 1.8 | .9641 | .9649 | .9656 | .9664 | .9671 | .9678 | .9686 | .9693 | .9699 | .9706 |
| 1.9 | .9713 | .9719 | .9726 | .9732 | .9738 | .9744 | .9750 | .9756 | .9761 | .9767 |
| 2.0 | .9772 | .9778 | .9783 | .9788 | .9793 | .9798 | .9803 | .9808 | .9812 | .9817 |
| 2.1 | .9821 | .9826 | .9830 | .9834 | .9838 | .9842 | .9846 | .9850 | .9854 | .9857 |
| 2.2 | .9861 | .9864 | .9868 | .9871 | .9875 | .9878 | .9881 | .9884 | .9887 | .9890 |
| 2.3 | .9893 | .9896 | .9898 | .9901 | .9904 | .9906 | .9909 | .9911 | .9913 | .9916 |
| 2.4 | .9918 | .9920 | .9922 | .9925 | .9927 | .9929 | .9931 | .9932 | .9934 | .9936 |
| 2.5 | .9938 | .9940 | .9941 | .9943 | .9945 | .9946 | .9948 | .9949 | .9951 | .9952 |
| 2.6 | .9953 | .9955 | .9956 | .9957 | .9959 | .9960 | .9961 | .9962 | .9963 | .9964 |
| 2.7 | .9965 | .9966 | .9967 | .9968 | .9969 | .9970 | .9971 | .9972 | .9973 | .9974 |
| 2.8 | .9974 | .9975 | .9976 | .9977 | .9977 | .9978 | .9979 | .9979 | .9980 | .9981 |
| 2.9 | .9981 | .9982 | .9982 | .9983 | .9984 | .9984 | .9985 | .9985 | .9986 | .9986 |
| 3.0 | .9987 | .9987 | .9987 | .9988 | .9988 | .9989 | .9989 | .9989 | .9990 | .9990 |
| 3.1 | .9990 | .9991 | .9991 | .9991 | .9992 | .9992 | .9992 | .9992 | .9993 | .9993 |
| 3.2 | .9993 | .9993 | .9994 | .9994 | .9994 | .9994 | .9994 | .9995 | .9995 | .9995 |
| 3.3 | .9995 | .9995 | .9995 | .9996 | .9996 | .9996 | .9996 | .9996 | .9996 | .9997 |
| 3.4 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9997 | .9998 |
| 3.5 | .9998 | .9998 | .9998 | .9998 | .9998 | .9998 | .9998 | .9998 | .9998 | .9998 |

# 3

## Extreme-value distributions

# Extreme Value Distributions

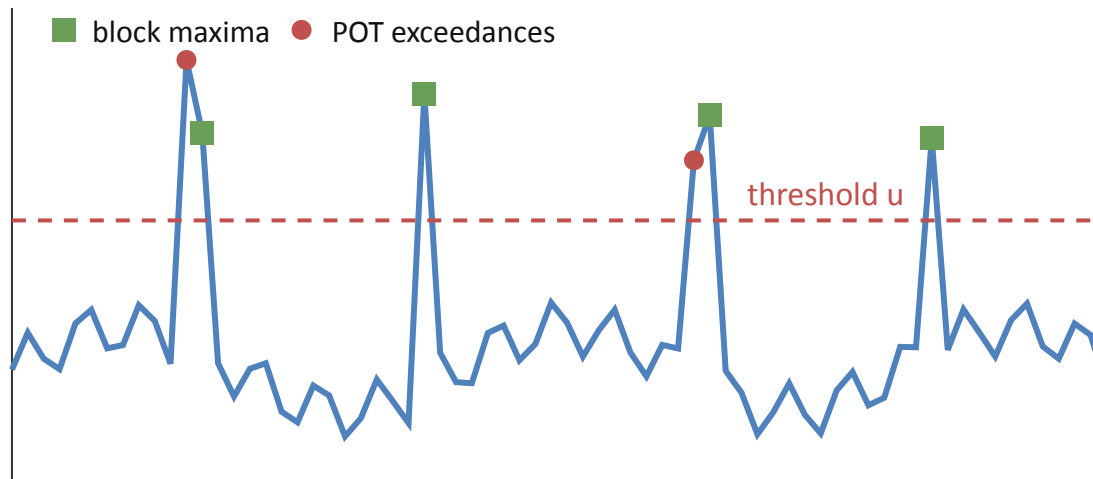
- $x_1, x_2, \dots, x_n$  random, independent variables with the same distribution.
- $x_1, x_2, \dots, x_n \rightarrow$  mean  $\mu$  & variance  $\sigma^2$
- **Question:** The extreme values of  $x_1, x_2, \dots, x_n$  has which distribution?
  - $\rightarrow$  Extreme value theory
  - $\rightarrow$  Block maxima and threshold maxima

# Extreme Value Distributions

## Block maxima & Threshold maxima:

- If  $X_n = \max(x_1, x_2, \dots, x_n)$ . Eg.  $X_n$  is annual  $TX_x$  or  $Rx1day \rightarrow X_n$ : block maximum.
- If  $Z_n$  are rainfall values of days in a year of which the daily rainfall amount  $\geq 20mm \rightarrow Z_n$  is a series of threshold maxima
- What are the distributions of  $X_n$  and  $Z_n$ ?

# Block maxima and peaks-over-threshold use different data from the same series



## Block maxima

One maximum per block; simple but discards other large events.

## POT

All exceedances above high threshold  $u$ ; more efficient but threshold/dependence choices matter.

# Extreme Value Distributions

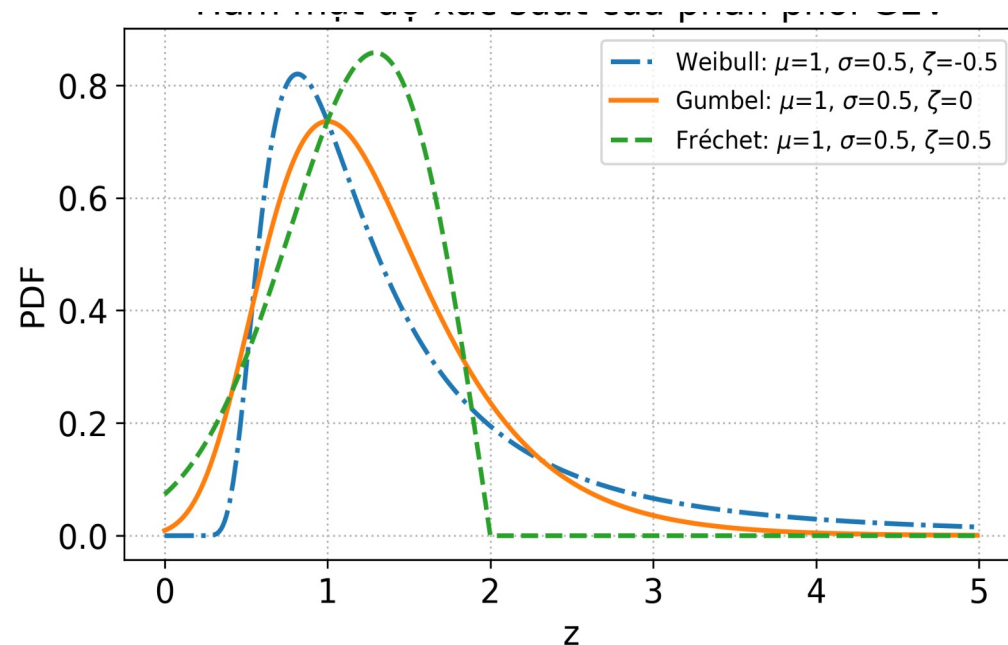
## Generalized Extreme Value (GEV) distribution

- The CDF of a block maxima  $z$  :

$$G(z) = \exp \left[ - \left\{ 1 + \xi \left( \frac{z - \mu}{\sigma} \right) \right\}^{-1/\xi} \right]$$

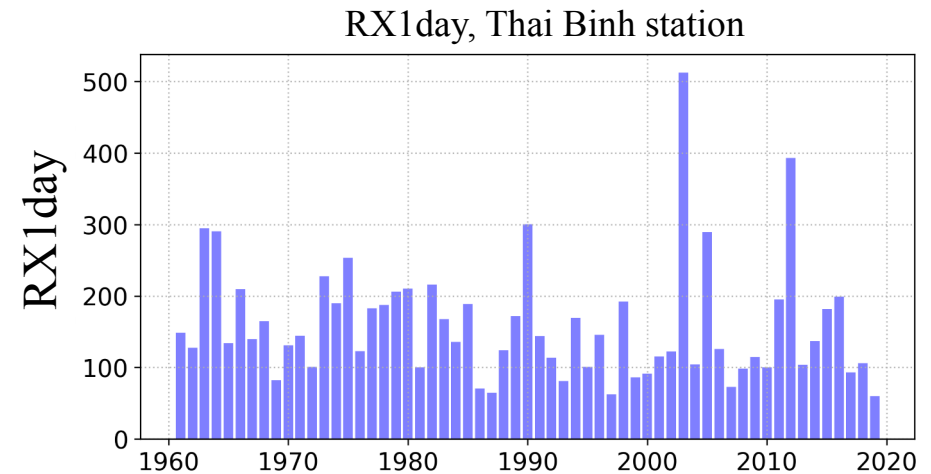
$G(z)$  has 3 parameters, including:

- location  $\mu$
- scale  $\sigma$
- shape  $\xi$ ;
  - $\xi=0 \rightarrow$  Gumbel distribution
  - $\xi>0 \rightarrow$  Fréchet distribution (bounded tail)
  - $\xi<0 \rightarrow$  Weibull distribution (heavy tail)



# Example

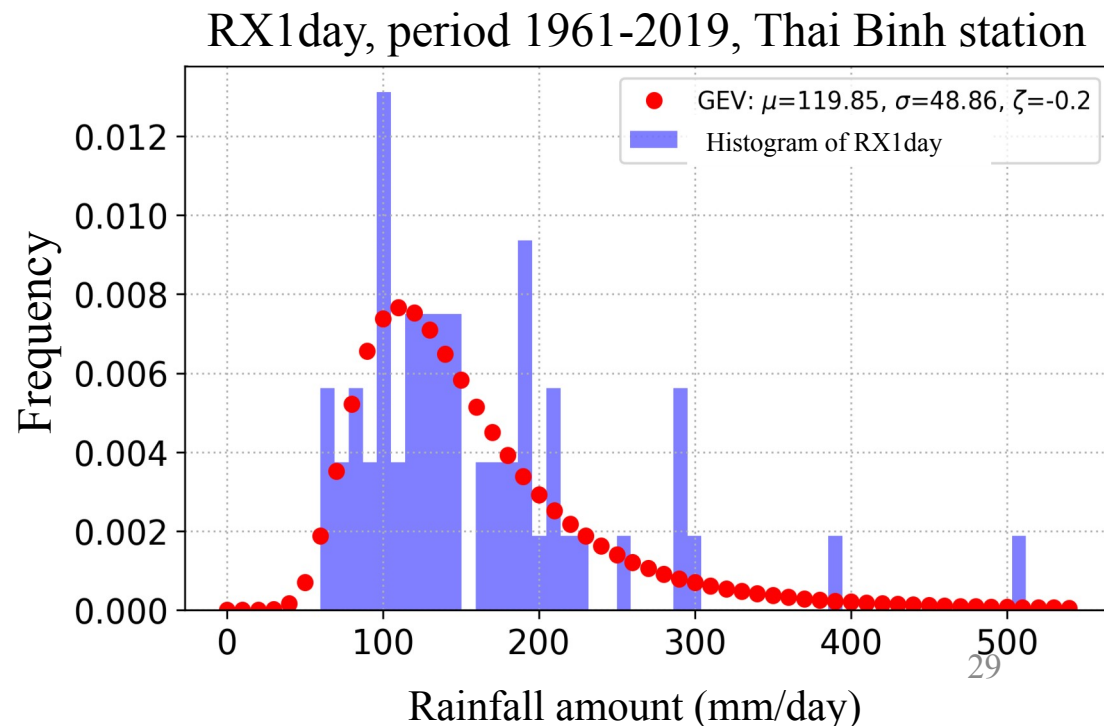
- Rx1day is the annual maximum daily rainfall, Thai Binh station, period 1961–2019 → a series of 59 values



From 59 RX1day values → the GEV distribution with 3 parameters:

$$\mu=119.85, \sigma=48.86, \text{ \& } \xi=-0.2$$

→ Rx1day → approximated by the Weibull distribution.



# Example

- Distributions of TXx & TNn at Thai Binh station

- TXx → GEV distribution with 3 parameters:

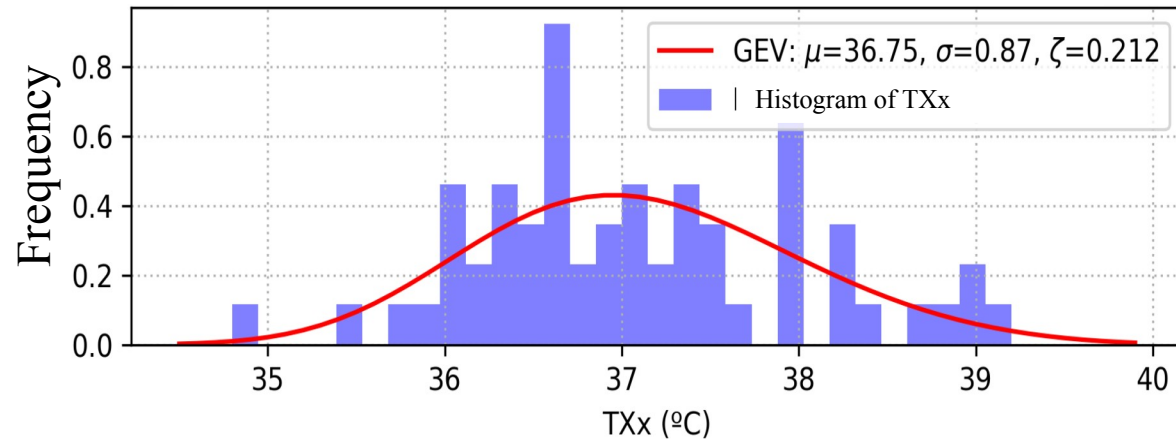
$$\mu=36.75, \sigma=0.87, \text{ \& } \xi=0.212$$

- TNn → GEV distribution with 3 parameters:

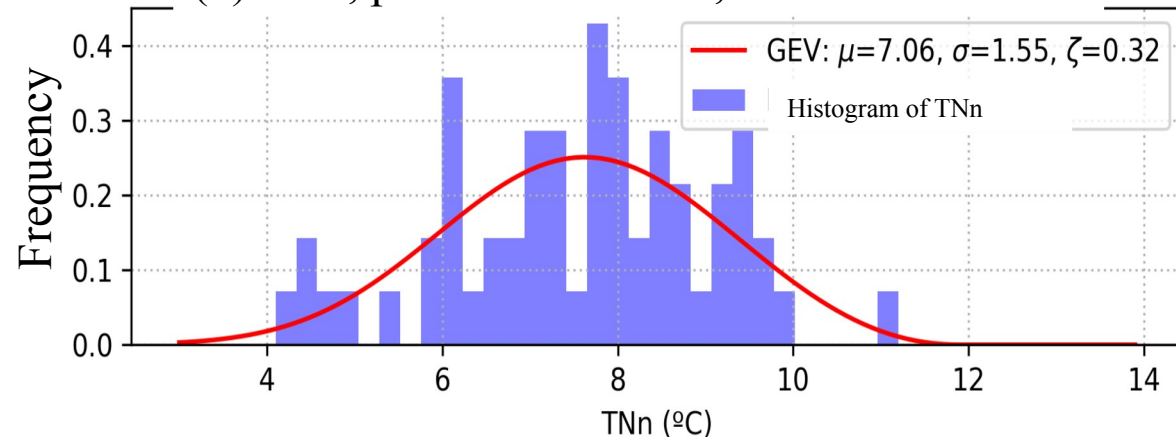
$$\mu=7.06, \sigma=1.55, \text{ \& } \xi=0.32$$

→ approximated by the Fréchet distribution

(a) TXX, period 1961-2019, Thai Binh station



(b) TNn, period 1961-2019, Thai Binh station



# Extreme Value Distributions

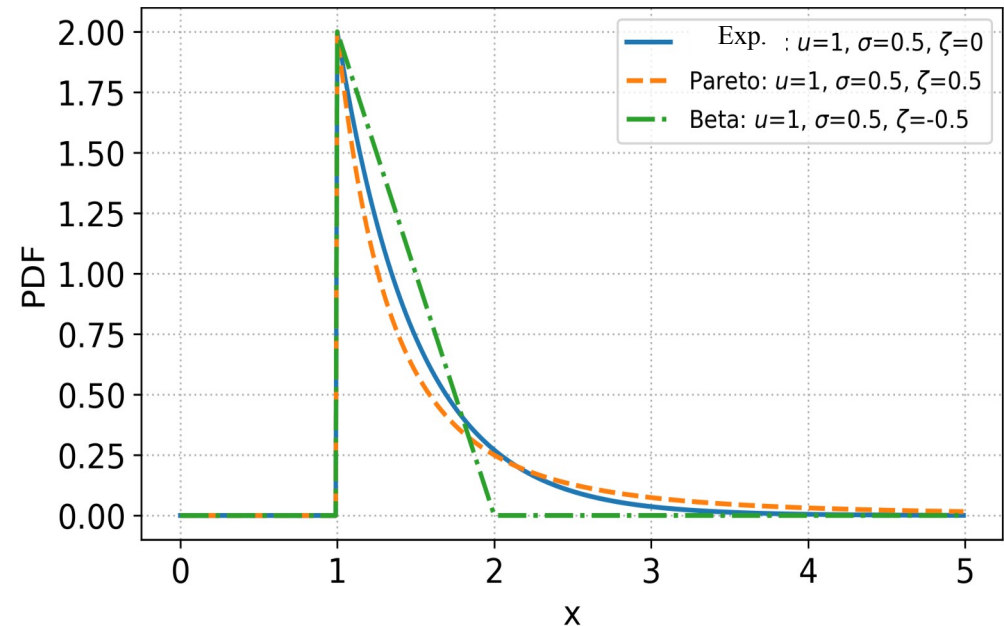
## Generalized Pareto (GP) distribution

GP distribution is often applied for threshold maxima

$H(x)$  has 3 parameters:

$$H(x) = 1 - \left[ 1 + \xi \left( \frac{x - u}{\sigma} \right) \right]^{-1/\xi}$$

- threshold  $\mu$
- scale  $\sigma$
- shape  $\xi$ 
  - $\xi > 0 \rightarrow$  Pareto distribution
  - $\xi < 0 \rightarrow$  Beta distribution
  - $\xi \rightarrow 0 \rightarrow$  exponential distribution



**80/20 rule (or Pareto principle):** in many events,  
80% of outcomes result from 20% of all causes

# Example

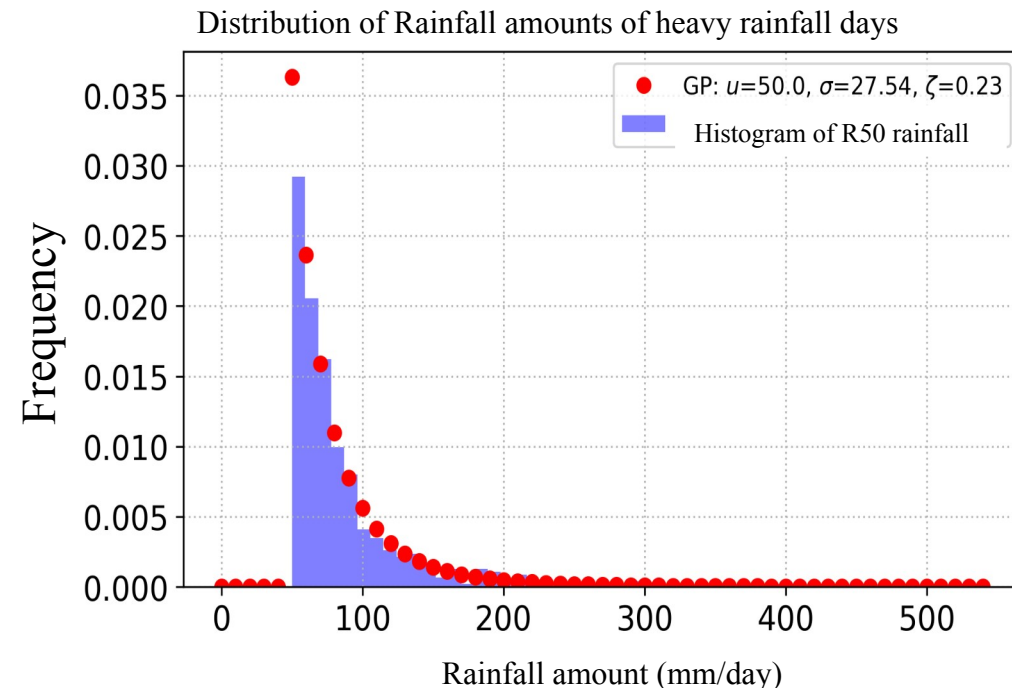
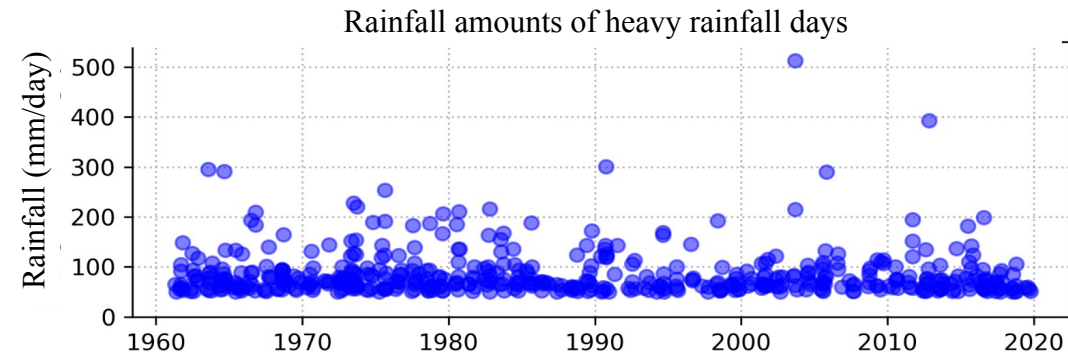
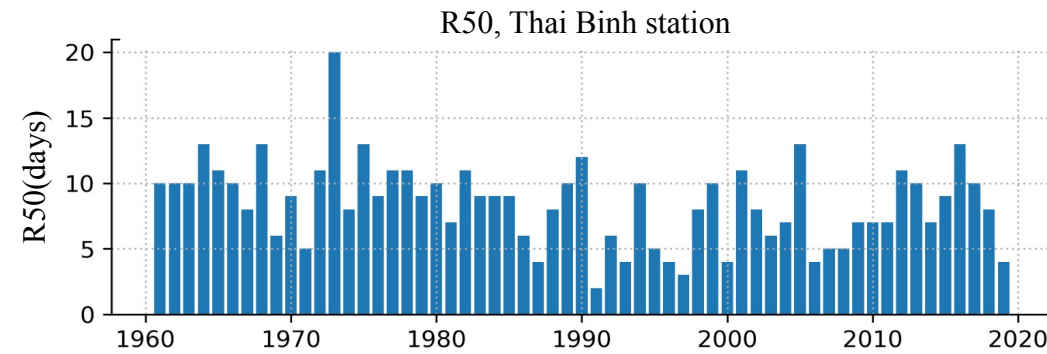
- ? Distribution of the rainfall amounts of heavy rainfall days, i.e. days with daily rainfall  $\geq 50\text{mm}$ , Thai Binh station, period 1961–2019

→ 500 days with heavy rainfall

- Rainfall values of 500 heavy rainfall days  
→ GP distribution with 3 parameters:

$\xi > 0 \rightarrow$  Pareto distribution.

$\mu=50, \sigma=27.54, \& \xi=0.23$



# Extreme Value Distributions

## Return level

- From the distribution  $\rightarrow$  can calculate the return level of an event
- $x_T$  is the value that is expected to be repeated on average one time per one period  $T$  (with a probability  $1/T$ )
- $T \rightarrow$  return period

- If  $x$  is block maxima,  $x_T$  satisfies:

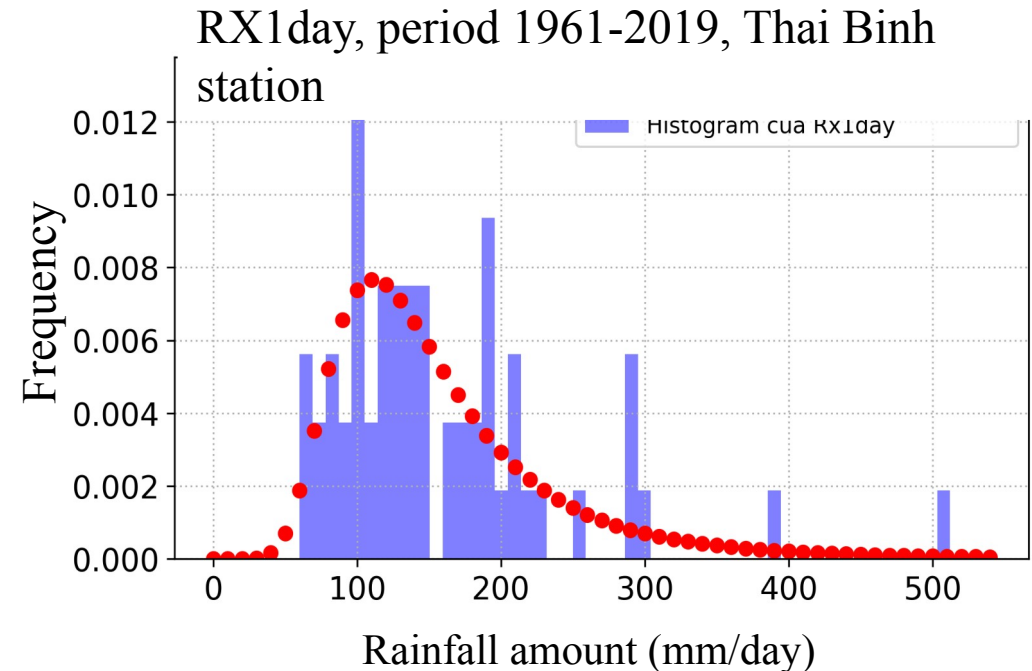
$$G(x_T) = 1 - 1/T$$

- If  $x$  is threshold maxima,  $x_T$  satisfies:

$$H(x_T) = 1 - 1/T$$

# Example

- GEV Weibull distribution for Rx1day, Thai Binh station, with 3 parameters:  $\mu=119,85$ ,  $\sigma=48,86$ , &  $\xi=-0,2$ .
- Calculate the return level of Rx1day for each period 2 years, 5 years, 10 years, 20 years!?



→ Solve the equation:

$$G(z) = \exp \left[ - \left\{ 1 + \xi \left( \frac{z - \mu}{\sigma} \right) \right\}^{-1/\xi} \right] = 1 - 1/T$$

for  $T=2, 5, 10, 20$  years

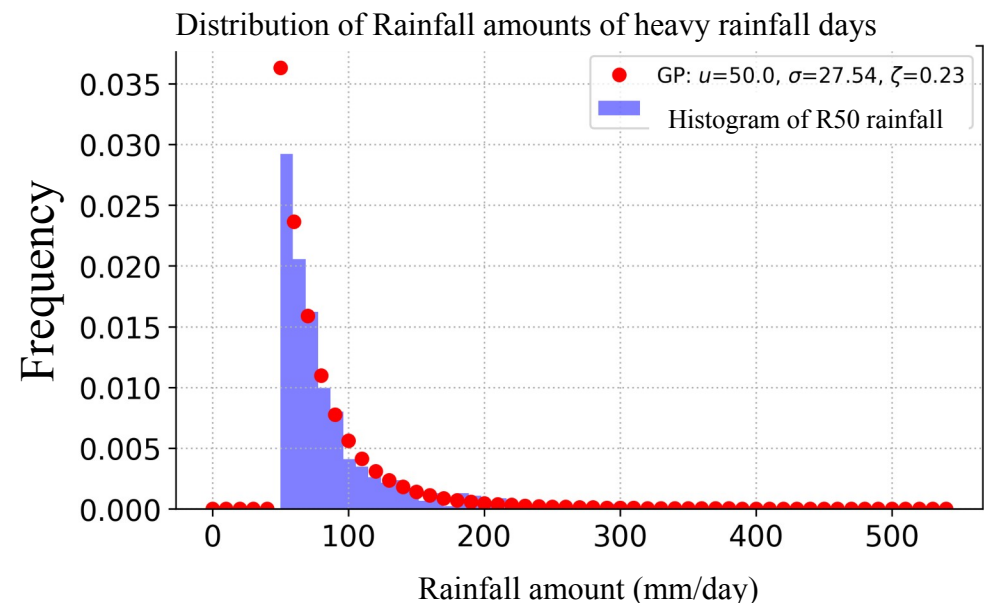
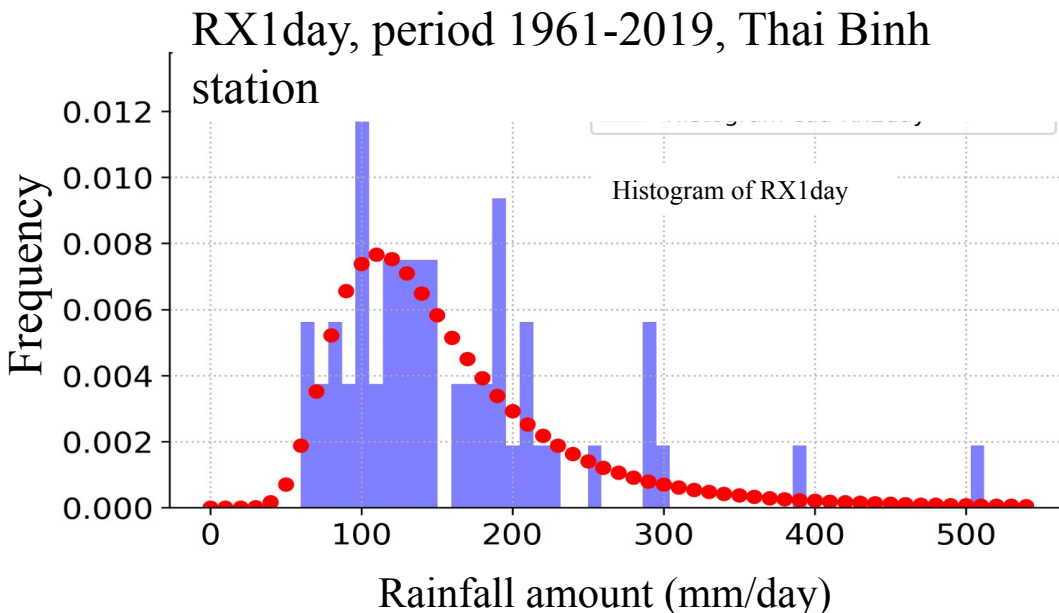
→ Results: the return levels are:

- ✓ 138.4 mm/day for 2-yr return period
- ✓ 205.3 mm/day for 5-yr return period
- ✓ 258.7 mm/day for 10-yr return period
- ✓ 318.0 mm/day for 20-yr return period

# Practice Python #12

Given the ThaiBinh rainfall data (daily, 1961–2019, [http://remosat.usth.edu.vn/~thanhnd/Download/dat\\_THAIBINH/R\\_THAIBINH\\_1961\\_2019.txt](http://remosat.usth.edu.vn/~thanhnd/Download/dat_THAIBINH/R_THAIBINH_1961_2019.txt))

1. Estimate RX1day for each year
2. Plot the distribution of RX1day
3. Calculate the return value of RX1day for the return period 5 yr, 10yr, 20 yr, 100 yr
4. **Do similarly for the rainfall amounts of the heavy rainfall days (days with rainfall  $\geq 50\text{mm}$ )**



# GEV

```
from scipy.stats import genextreme as gev
```

```
# Determine the parameters of the GEV distribution
```

```
shape, loc, scale = gev.fit(Rx1day) # estimate the 3 parameters of the GEV
```

```
# Estimate the return values corresponding to the return periods 5 years
```

```
x5=gev.ppf((1-1/5.),shape,loc,scale)
```

```

import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import genextreme as gev
r2rain=np.loadtxt("R_THAIBINH_1961_2019.txt",comments="#")
nyr=int(len(r2rain[:,1])/31)      # number of years
# reformat to 3D: r3T2m[year,dayofmonth,month]
r3rain=np.reshape(r2rain[:,1:13],(nyr,31,12))
# Calculate Rx1day for each year
Rx1day=np.zeros(nyr)
iyr=0
while iyr < nyr:
    imn=0
    while imn <12:
        idy=0
        while idy<31:
            if r3rain[iyr,idy,imn]>= Rx1day[iyr]:
                Rx1day[iyr]=r3rain[iyr,idy,imn]
            idy=idy+1
            imn=imn+1
        iyr=iyr+1
# Determine the parameters of the GEV distribution
shape, loc, scale = gev.fit(Rx1day) # estimate the 3 parameters of the GEV
x = np.arange(0,550,10)
y = gev.pdf(x, shape, loc, scale)
# Estimate the return values corresponding to the return periods 5, 10, 20, 100 yrs
x5=gev.ppf((1-1/5.),shape,loc,scale)
x10=gev.ppf((1-1/10.),shape,loc,scale)
x20=gev.ppf((1-1/20.),shape,loc,scale)
x100=gev.ppf((1-1/100.),shape,loc,scale)
print(x5,x10,x20,x100)

```

```

## Plot
fig, ax1 = plt.subplots(figsize=(7,4))
ax1.hist(Rx1day,bins=50,density=True, histtype='stepfilled',color='blue',\
        alpha=0.5,label='Histogram of Rx1day')
plt.plot(x,y,'ro',label='GEV: $\mu$='+str(round(loc,2))+',\
        $\sigma$='+str(round(scale,2))+', $\zeta$='+str(round(shape,3)))
plt.legend(loc='upper right')
ax1.set(title='Rx1day 1961-2019, ThaiBinh station',\
        xlabel='Daily Rainfall (mm/day)',ylabel='Frequency')

```

205.2976342634028 258.67226113738667 317.96364801694955 488.3353055305761

